HARD · INFORMATION AND IDEAS · ANSWER KEY

Information and Ideas

01

10 answers · Rationale-rich · Teach from doc

Q1

C

C. iron from HTC dust is 90%.

Choice C is the best answer because it most effectively completes the example regarding the ablation rate of iron. The table shows the ablation rates for three elements—iron, potassium, and sodium—found in cosmic dust that comes from one of four sources. The text says that the ablation rate for a given element in slower-moving SPC and AST dust was lower than the ablation rate for that same element in faster-moving HTC or OCC dust. The text then presents the first part of an example of this pattern, describing an ablation rate of 28% for iron in AST dust. The information that iron from HTC dust had an ablation rate of 90% is therefore the most effective way to complete this example—the comparison of a relatively low ablation rate for iron in slower-moving AST dust with a relatively high ablation rate for iron in faster-moving HTC dust illustrates the tendency of ablation rates for a given element to be lower in slower-moving dust than in faster-moving dust.

Choice A is incorrect because the text indicates that SPC dust, like AST dust, moves relatively slowly; a comparison of the ablation rates of iron from two slower-moving dust sources could not be an example of the difference between ablation rates in slower-moving dust and faster-moving dust, which is the pattern that the example is supposed to illustrate. Choice B is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Choice D is incorrect because the example in the text is supposed to illustrate the difference in the ablation rates of the same element from slower-moving dust and faster-moving dust, and the first part of the example provides data about the ablation rate of iron, which means the second part of the example must also be about the ablation rate of iron, not the ablation rate of sodium. Additionally, any ablation rate from AST dust would be ineffective in this example since AST dust is referenced in the first part of the example and thus additional data focused on AST dust would not illustrate a variation across dust types.

Q2 D

D. Mars's crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

Choice D is the best answer. Cooling magma would create basalt, but "a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica." Since Mars's crust does contain large amounts of silica, it is unlikely that Mars's crust was formed exclusively by cooling magma. Therefore, there were likely other major geological events that created the high silica concentrations.

Choice A is incorrect. Although the passage discusses these two methods of collecting data about Mars's crust, it never compares their reliability, so there's no basis for this inference. Choice B is incorrect. The passage never mentions anything about the crusts of other planets, so there's no basis for this inference. Choice C is incorrect. The passage never mentions Earth's crust, so there's no basis for this inference.

Q3

C

C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.

Choice C is the best answer because it states a conclusion the researchers likely agree with, given the details in the text. The text explains that a biosignature gas is a gas that can be used as an indicator that a planet harbors some form of life and some astronomers have proposed that NH_3 could serve as a biosignature gas. The researchers evaluating this claim found that the atmosphere of rocky planets would be unlikely to reach "detectably high levels" of NH_3 without biological activity, which would support the proposal of NH_3 serving as a biosignature gas. However, the text also states that mini-Neptune planets can produce NH_3 in the absence of biological activity. Thus, the text is structured to lead to the conclusion that detectable levels of NH_3 in the atmospheres of rocky planets could constitute a biosignature, but that is not the case for detectable levels of the gas in the atmospheres of mini-Neptune planets.

Choice A is incorrect because the text indicates that biological activity likely accounts for detectable levels of NH_3 in the atmospheres of rocky planets but mini-Neptune planets can have detectable levels of NH_3 in their atmospheres in the absence of biological activity. Therefore, both rocky planets and mini-Neptune planets can have detectable levels of atmospheric NH_3. Choice B is incorrect because the text states that for NH_3 to reach detectable levels in the atmospheres of rocky planets likely means they harbor biological activity, meaning that rocky planets with detectable NH_3 usually harbor biological activity. However, that does not entail that every rocky planet with biological activity will have detectable levels of NH_3 in their atmospheres. Choice D is incorrect because the text claims only that some astronomers have proposed using NH_3 as a biosignature gas without mentioning a minimum concentration of atmospheric NH_3 that must be met for it to function as a biosignature gas.

Q4

A

A. national governments of countries in Region 1 experienced declines in efficiency in the period from 2000 to 2014, relative to the period from 1970 to 2000.

Choice A is the best answer because it most effectively uses data from the graph to complete the statement about Mahtta et al.'s proposal regarding factors that affect urban land expansion (ULE). According to the text, ULE is influenced by urban population growth and by gross domestic product (GDP) growth per capita. Reasoning that efficient national governments provide urban services and infrastructure needed to attract economic investment, Mahtta et al. suggest that, as governments become more efficient at providing urban services and infrastructure, GDP growth per capita will account for more ULE and urban population growth will account for less. But according to the graph, Region 1 saw an increase in the percentage attributed to urban population growth from 1970–2000 (between 60 and 65%) to 2000–2014 (between 70 and 75%) and a decrease in the percentage attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%). Because the percentage attributed to GDP growth per capita decreased (the opposite of what Mahtta et al. claimed would happen if the governments had become more efficient), the data suggest that the governments of Region 1 became less efficient at providing urban services and infrastructure over that period.

Choice B is incorrect. Neither the graph nor the text gives the regions' relative levels of economic growth or what effect Mahtta et al. would expect such growth to have. Furthermore, Mahtta et al.'s proposal suggests that Region 1's decline in the percentage of ULE attributed to GDP growth per capita from 1970–2000 (between 35 and 40%) to 2000–2014 (about 25%) would suggest decreasing, not increasing, government efficiency over this time. Choice C is incorrect. Neither the text nor the graph provides information about the relative efficiencies of different governments in Region 2. Choice D is incorrect. Mahtta et al.'s proposal suggests that more efficient governments will have a higher percentage of their ULE driven by GDP growth per capita and a lower percentage driven by urban population growth. For Region 2, the percentage of ULE attributed to GDP growth per capita increased from 1970–2000 (between 10 and 15%) to 2000–2014 (between 45 and 50%), but the opposite is true for Region 1, which saw the percentage of ULE attributed to GDP growth per capita decline over the same period. Thus, whereas the data suggest governments in Region 2 became more efficient, the data for Region 1 suggest that those governments became less efficient, not more.

Q5 D

D. the family's division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztecan tribes.

Choice D is the best answer because it most logically completes the discussion of Uto-Aztecan languages. The text explains that the northern and southern branches of the Uto-Aztecan language family descended from a single language (believed to have originated in what is now the US Southwest), resulting in similarities across the family's languages; however, the branches don't have similar vocabulary for maize, even though maize has been cultivated by all Uto-Aztecan tribes. The text also indicates that maize originated in Mexico and spread northward into what is now the US Southwest—the area where the Uto-Aztecan language family originated. It follows, then, that the language family had already divided into northern and southern branches before maize reached that area; if maize had been present before the division occurred, the family's origin language would have had terminology for it that likely would have been reflected in the branches, meaning they would have had similar vocabulary for maize. If maize arrived after the division occurred, however, the tribes in the two regions likely would have developed vocabulary pertaining to maize separately, at the times when they acquired the crop.

Choice A is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztecan language family, and referring only to how some Uto-Aztecan tribes obtained maize wouldn't directly address the role of language. Moreover, if northern Uto-Aztecan tribes had acquired maize from a southern Uto-Aztecan tribe, it's reasonable to assume that the northern tribes might have also picked up southern Uto-Aztecan terminology for maize in that exchange. Choice B is incorrect because the text discusses the fact that the northern and southern branches of the Uto-Aztecan language family don't have shared vocabulary pertaining to maize, not the idea that there are variations in such vocabulary within each branch—that is, the text focuses on differences between the two branches, not on differences between languages within a branch. Choice C is incorrect because the text focuses on vocabulary pertaining to maize in the branches of the Uto-Aztecan language family, and referring only to the timing and source of maize acquisition wouldn't directly address the role of language. Furthermore, the text implies that southern Uto-Aztecan tribes probably acquired maize before the northern tribes did, given the evidence that maize originated in Mexico—the location of the best-known representative of the southern branch of the Uto-Aztecan language family—before spreading to the north.

Q6 C

C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.

Choice C is the best answer. The text argues that disco is "far less superficial" than its popular perception might indicate, and that love and heartbreak are "subjects hardly unique to disco."

Choice A is incorrect. This choice conflicts with the text, which says that scholars argue that disco "is far less superficial than many people believe." Choice B is incorrect. This choice says the opposite of what the text suggests. The writer argues that the genre is not as superficial as commonly believed, but that it always reflected "concerns about community and identity." Choice D is incorrect. The text doesn't support this choice. There's nothing in the text about disco giving rise to an enduring Black women's musical tradition.

Q7 D

D. The cost to rural entrepreneurs to bring their goods to towns with marketplaces is high but largely independent of the number of goods they bring.

Choice D is the best answer because it presents a finding that, if true, would most directly support Barboza and Trejos's claim that rural female entrepreneurs who have received small loans from ALSOL are strategic in selling their goods less frequently than they could, even if it means missing payments. The text explains that borrowers in the ALSOL program use proceeds from their businesses to repay loans in equal weekly payments, with almost no penalty for missed payments. According to the text, Barboza and Trejos found that rural borrowers miss weekly payments in part because they don't sell their goods as often as they could, a move the researchers claim allows the entrepreneurs to help increase profits for the goods they sell. Finding that the cost of bringing goods to towns with marketplaces is high for rural entrepreneurs but is largely independent of how many goods are brought would support the researchers' claim: traveling to marketplaces less frequently would mean that a rural entrepreneur spends less on travel overall, and taking a large load of goods to a marketplace for essentially the same cost as taking a small load would allow the entrepreneur to more substantially offset the cost of travel with greater overall sales at the marketplace, resulting in more profit per good sold—even if those profits are earned less frequently and don't support weekly loan payments.

Choice A is incorrect because the finding that many marketplaces require entrepreneurs to pay the operators of the marketplace a fixed percentage of proceeds to be able to sell goods there wouldn't explain why rural entrepreneurs strategically choose to sell their goods less frequently than they could in order to increase their profits per unit sold. With a fixed percentage of proceeds due to operators, the amount entrepreneurs have to pay operators would also be fixed regardless of frequency of selling. Choice B is incorrect because the finding that rural entrepreneurs can usually sell their goods for higher prices in cities than in their local areas but also face higher competition to sell goods in cities wouldn't explain why rural entrepreneurs strategically choose to sell their goods less frequently than they could in order to increase their profits per unit sold. This is because both the higher prices and higher competition in cities would be stable factors—meaning there would be no clear reason for the rural entrepreneurs not to take every available chance to sell their goods in cities and to instead sell their goods in cities only sometimes. Choice C is incorrect because the finding that rural entrepreneurs have lower costs and thus tend to require smaller initial loans than urban entrepreneurs do has no bearing on rural borrowers strategically choosing to sell their goods less frequently than they could specifically to increase their profits per unit sold. The cost of producing goods doesn't depend on the frequency with which an entrepreneur sells those goods, so lower frequency alone wouldn't affect profits, and the initial loan amount is set and has nothing to do with how much profit is earned from each sale.

Q8

A

A. risk judging Tanner's painting by standards that may not be historically appropriate.

Choice A is the best answer. It most logically completes the text. The text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. This suggests that scholars who attribute those modern values to Tanner's painting are risking judging the painting by standards that are not historically accurate.

Choice B is incorrect. It doesn't logically complete the text. The text argues that Tanner AND his contemporaries may have been unfamiliar with modern views. It never suggests that Tanner's views were different from his contemporaries' views. Choice C is incorrect. It doesn't logically complete the text. The text never suggests that scholars should analyze Tanner's political activity instead of his painting. Rather, the text argues that Tanner and his contemporaries may have been unfamiliar with modern beliefs and values. Choice D is incorrect. It doesn't logically complete the text. The text never suggests that Tanner wanted to critique his contemporaries with his painting. Rather, the text argues that Tanner AND his contemporaries may have been unfamiliar with modern beliefs and values.

Q9 D

D. He is overly protective of Evelina, who would likely benefit from a greater variety of experiences than she has had thus far.

Choice D is the best answer because it presents a statement about Reverend Villars that Lady Howard would most likely agree with based on the text. In the text, Lady Howard urges Reverend Villars to allow his ward, Evelina, to join the Mirvans on a trip to London. Her first words after describing the Mirvans' wish—"Do not start at this proposal"—suggest that she believes Villars will be alarmed by the idea, and she goes on to assure him that this alarm is unwarranted because Evelina will benefit from visiting London. Lady Howard supports her claim by arguing that excessive isolation from the world leads young people to entertain unrealistic, idealized views of it. On the other hand, she suggests, giving young people appropriate access to the world when they, like Evelina, are ready can help them develop a more balanced view of the world and its moral complexities. These words indicate that Lady Howard believes Evelina risks becoming too sheltered because of her guardian's protectiveness and that she would benefit from exposure to a wider range of experiences than she has had thus far.

Choice A is incorrect because while the text does suggest that Lady Howard believes Reverend Villars has sheltered Evelina too much, it doesn't specifically mention shielding her from the influence of other young people. Lady Howard refers more broadly to getting more exposure to the world and doesn't focus on peer relationships. Additionally, the text doesn't indicate whether Lady Howard finds other aspects of how Villars has raised Evelina to be exemplary. Choice B is incorrect because the text doesn't suggest that Lady Howard believes Reverend Villars's view of the world is idealistic or that he has imparted an idealistic view of the world to Evelina. While Lady Howard does warn that young people who are too sheltered may develop romantic, idealized views of the world, she presents this as a general principle about sheltered youth, not as a specific criticism of Villars for having transmitted a naïve outlook to Evelina. Choice C is incorrect because the text doesn't mention Reverend Villars being specifically concerned about protecting Evelina from unscrupulous people or suggest that he is exceptionally wary of the Mirvans. While Lady Howard's words certainly anticipate some resistance from Villars, the text doesn't specify the nature of his concerns beyond the general desire to protect Evelina from the world.

Q10 A

A. broccoli grown in soil containing mycorrhizal fungi had a slightly higher average mass than broccoli grown in soil that had been treated to kill fungi.

Choice A is the best answer because it most effectively uses data from the table to complete the statement. The text explains that mycorrhizal hosts are plants that benefit from the presence of mycorrhizal fungi in the soil and that some such plants produce more mass when grown in the presence of these fungi, while for nonmycorrhizal species the fungi either have no effect or may be harmful. The experiment included two mycorrhizal hosts (corn and marigold) and one nonmycorrhizal species (broccoli). Given the claim in the text that nonmycorrhizal species will see either no difference or a decrease in mass when exposed to mycorrhizal fungi, the student would likely have been surprised by the higher average mass for broccoli grown in the presence of the fungi than the broccoli grown in the soil treated to kill fungi.

Choice B is incorrect. Although this choice accurately describes the corn data from the table, the fact that the mycorrhizal host corn is more massive in the presence of the fungi likely fits with what the student expected and would therefore not be surprising. Choice C is incorrect. Although this choice accurately describes the marigold data from the table, the fact that the mycorrhizal host marigold is more massive in the presence of the fungi is likely what the student expected and thus would not be surprising. Choice D is incorrect because it does not accurately represent the data in the table—when grown in soil treated to kill fungi, corn had an average mass of 3.8 g while broccoli had an average mass of 7g—and because making comparisons among the plants in the no-fungi condition, by itself, does not provide a basis to compare the average mass of mycorrhizal hosts and nonmycorrhizal species grown in the presence of the fungi with those grown in the soil treated to kill fungi.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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